

CNL

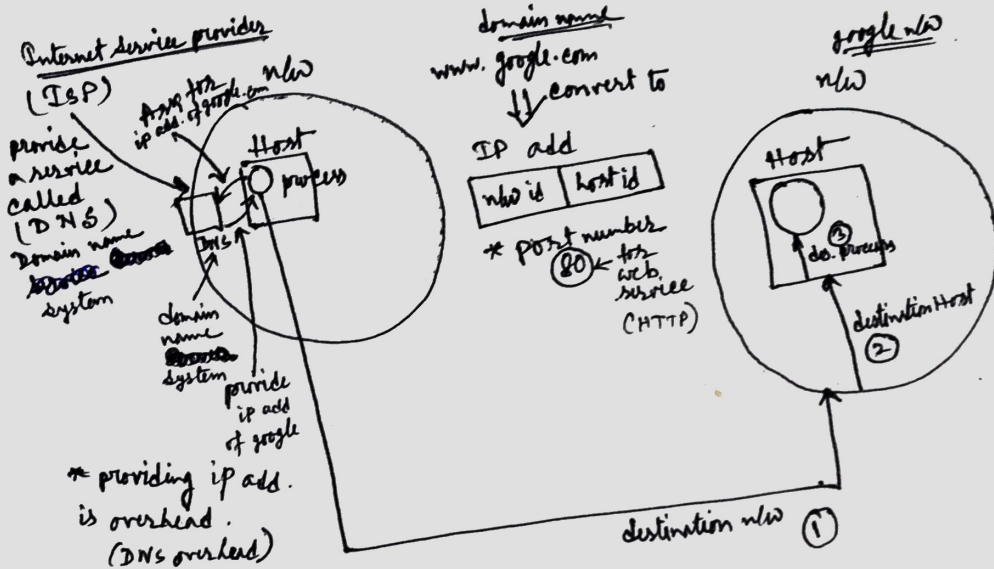
1 IP address, Subnetting, Supernetting

1.1

Intro. to computer n/w & IP address

* for every process there have an ip port number diff

ex. ~~21~~ $\frac{21}{25}$
FTP $\Rightarrow$ 21
SMTP $\Rightarrow$ 25 etc.



IP add.

$\Rightarrow$ Number System

- * **Unary NS** $\Rightarrow$ 0, 1, 00, 000
- * **decimal NS** $\Rightarrow$ 0 - 9
- * **Binary NS** $\Rightarrow$ 0, 1 (Base is 2)
If Base is n then we have 0, 1, 2, ..., (n-1) number
- * **Octal NS** $\Rightarrow$ 0, 1, 2, ..., 7
- * **Hexadecimal NS** $\Rightarrow$ 0, 1, 2, ..., 9, A, B, ..., F

Binary NS is most popular in CS

- $2^1 - 2$
- $2^2 - 4$
- $2^3 - 8$
- $\vdots$
- $2^{10} - 1024$
- $K - 2^{10}$
- $M - 2^{20}$ (Mega)
- $G - 2^{30}$ (Giga)
- $T - 2^{40}$ (Tera)

$\Rightarrow$ 1 Bit (we have 1 bit)

$\Rightarrow$ do how many numbers are possible in 1 bit?

1 Bit

0
1

2 Bit

number space
00 - 0
01 - 1
10 - 2
11 - 3

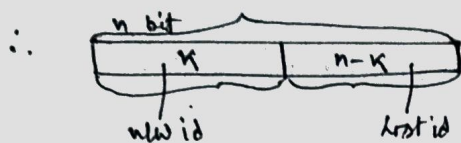
if i choose first bit, then i can divide entire space into two part

3 Bit	if i choose 1 bit	2 ¹ part (space)
000 - 0	2 - 2 ²	"
001 - 1	3 - 2 ³	"
010 - 2		
011 - 3		
100 - 4		
101 - 5		
110 - 6		
111 - 7		

do if i have 'n' bit number then 2ⁿ number space are present

⇒ Now if we choose n bit then we get 2^n part from 2^n numbers
 do ~~what~~ ^{how many} the numbers are contain in each part?

1 part — $\frac{2^n}{2^n} = 2^{n-n}$ numbers
 contain



⇒ so if i choose n -bit we get 2^n network & size of each network is 2^{n-k}

IP add

32-bits

so in 32 bit, the numbers of ip address are possible = 2^{32}

Initially computer n divided into 2 parts:

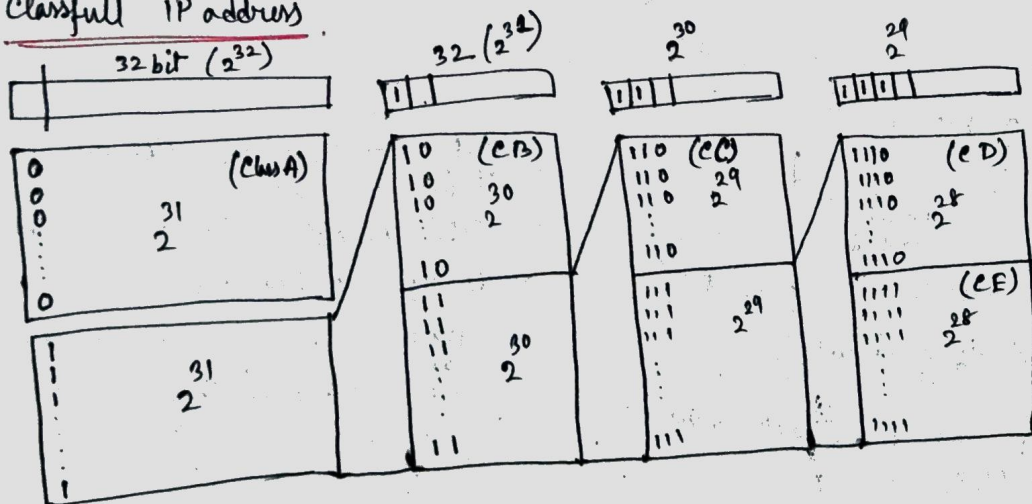


$2^8 \Rightarrow 256$ n

$2^{24} = 16$ million host

But in today's life only 256 n are not enough, so net we can use classfull IP addressing.

Classfull IP address

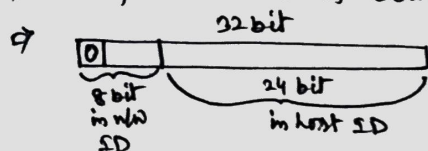


prefix code

0 — class A
 1 0 " B
 1 1 0 " C
 1 1 1 0 " D
 1 1 1 1 " E

class A

⇒ No. of IP address in class A is 2^31 IP add.



→ So in n/w ID first bit is fixed as 0, so from remaining bit (7 bit) how many n/w of class A are present?

$2^7 = 128$ n/w of class A are present
126 n/w (practically)

⇒ How many IP add. are present in each n/w?

$\frac{2^{24}}{2} = 16$ million IP add. are present in each n/w.

⇒ Range of IP add in class A (Only for the octet (i.e. n/w ID)) ⇒ 1-126

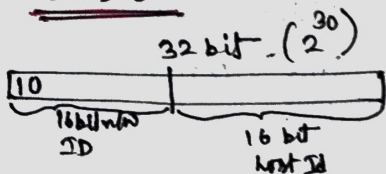
not used X $\begin{array}{l} 00000000 - 0 \\ 00000001 - 1 \\ 00000010 - 2 \end{array}$ } 126 n/w

$$\left[\begin{array}{l} 1 - 1 \\ 11 - 3 \\ 111 - 7 \end{array} \right] [2^3 - 1 = 7]$$

not used X $01111111 - 127$ } $[2^7 - 1 = 127]$ loop back IP add.

(*) Very big organization (like NASA, PENTAGON) buy it, but small organization can't afford it, so they can choose class B.

class B



⇒ How many n/w ID are present?

$2^{14} \approx 16$ K (Nearly)

⇒ How many IP add. in each n/w?

$\frac{2^{16}}{2} = 64$ K.

Range 128 - 191 } $\left(\begin{array}{l} 6 \\ 2^6 = 64 \end{array} \right)$ 64 n/w (for first octet)
 $\begin{array}{l} 10000000 - 128 \\ 00000001 - 129 \\ 00000010 - 130 \end{array}$ } (6 n/w)
 $10111111 - 191 (128 + 63)$

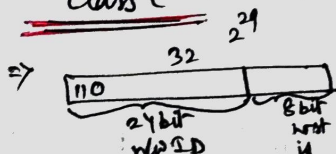
⇒ But we have another octet, & it give 2^8 n/w

⇒ So total $64 \times 2^8 = 2^{14}$ n/w are available
 so it represent like this:

$\left. \begin{array}{l} 128.0 \\ 128.1 \\ 128.2 \\ \vdots \\ 128.255 \end{array} \right\}$ So every no. are capable of presenting 255 n/w. & each n/w contain 2^{16} IP add.

$\left. \begin{array}{l} 191.0 \\ 191.1 \\ \vdots \\ 191.255 \end{array} \right\}$

Class C



$\Rightarrow 2^{24} \approx 2 \text{ million [networks]}$

$\Rightarrow 2^8 \text{ IP add | network}$

Range

110 0 0 0 0 0	- 192	} 32 network (for first octet) (5 one) (192 + 31)
0 0 0 0 1	- 193	
0 0 0 1 0	- 194	
...	...	
1 1 1 1 1	- 223	

$\Rightarrow$ So total network $2^8 \times 2^{24}$

- 192.0.0
- 192.0.1
- 192.0.255
- 192.1.0
- 192.1.1
- 192.255.255
- host 223.255.255

Class D & Class E

$\Rightarrow$ These two classes not divide into network & host part

Class D (It used for multicasting)

IP add. = 2^{28}

Range (224-239)

1110 - - - -
0 0 0 0 $\rightarrow$ 224

1110 - - - -
1 1 1 1 $\rightarrow$ 239

Class E

IP add. = 2^{28}

Range (240-255)

1111 - - - -
0 0 0 0 $\rightarrow$ 240

1111 - - - -
1 1 1 1 $\rightarrow$ 255

$\Rightarrow$ These two networks are not used for practical purposes.

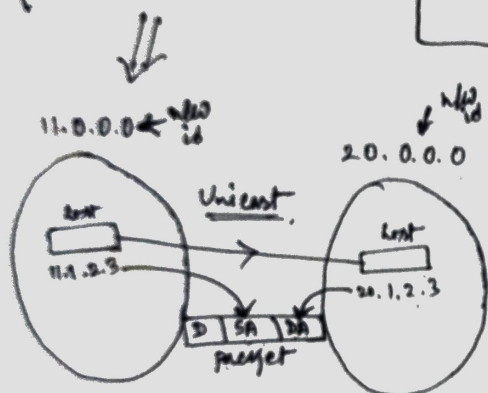
(L2) Types of Casting (Unicast, limited Broadcast, Directed Broadcast)

⇒ Now we see how to send a packet from one host to other host.
 ⇒ There is two ways. It called casting.

Casting

Unicast

(From one host to another particular host)



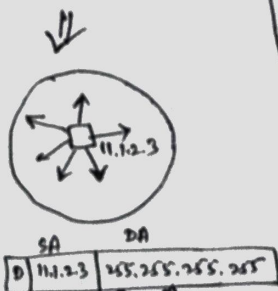
$NID = NID$, $HSD(All\ 0's)$
 $LBA = (255.255.255.255)$ (All 1's (1111))
 $DBA = NID, HSD(All\ 1's)$

Broadcast

(Sending a packet from one host to many host.)

Limited Broadcast

(A host send the msg. to all other nodes in same n/w)

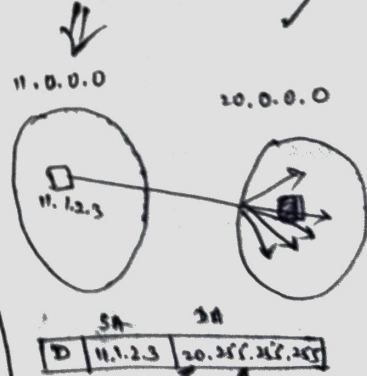


So, This IP add. is never be assign to any of the host, because it is a LBA.

Limited Broadcast add
 255.255.255.255

Directed Broadcast

(A host in a n/w send a msg to all other host in same other n/w.)



So, This IP add. is never be assign to any of the host, because it is a DBA.

Directed Broadcast address
 20.255.255.255
 That n/w ID

$$CA = 1 - 126$$

$$CB = 128 - 191$$

$$CE = 192 - 223$$

IP	NID	DBA	LBA
CA 1.2.3.4 10.15.20.60	$\frac{1.0.0.0}{NID} \quad \frac{h.id}{h.id}$ 10.0.0.0	1.255.255.255 10.255.255.255	255.255.255.255 255.255.255.255
CB 130.1.2.2 150.0.150.150	$\frac{130.1.0.0}{NID} \quad \frac{h.id}{h.id}$ 150.0.0.0	130.1.255.255 150.0.255.255	255.255.255.255 255.255.255.255
CE 200.1.10.100 220.15.1.10	$\frac{200.1.10.0}{NID} \quad \frac{h.id}{h.id}$ 220.15.1.0	200.1.10.255 220.15.1.255	" "
250.0.1.2 300.1.2.3	X X	X X	X X

L3 Subnets, Subnet Mask, Routing

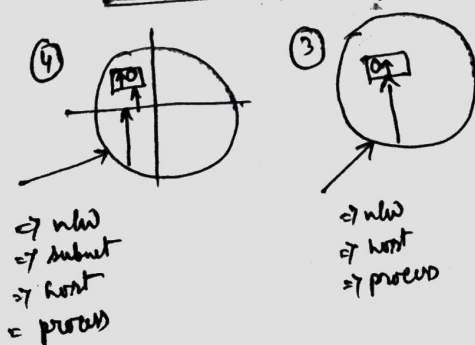
⇒ In class A ⇒ 16 million host

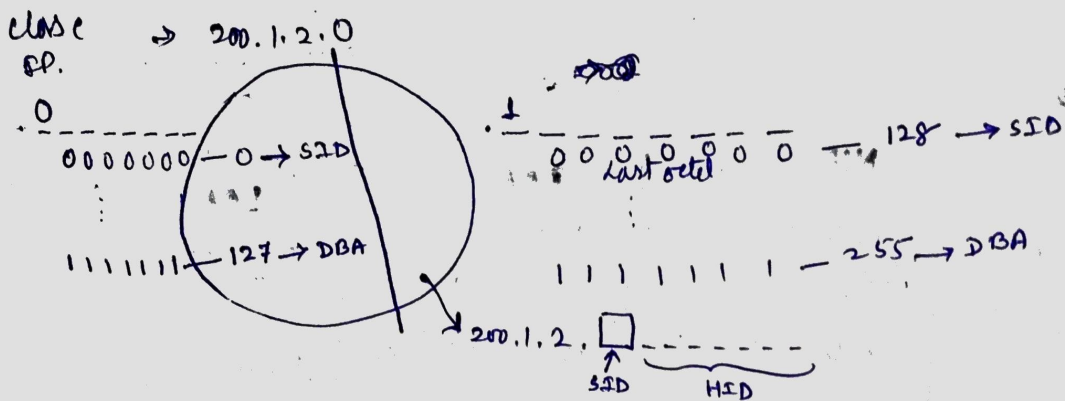
In class B ⇒ 16K "

In class C ⇒ 2⁸ "

⇒ So in class A or B it is very difficult to manage this much of host in a single net. (maintenance & security).

⇒ disadvantage for subnetting





$\Rightarrow$ Want to divide this nlw into two part.

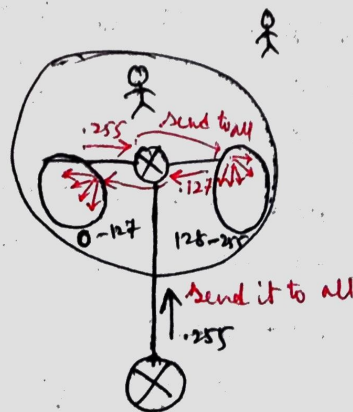
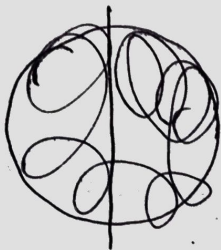
$\Rightarrow$ So we can choose first bit of last octet.

$\Rightarrow$ Now confusion is if we choose $200.1.2.0$ ~~then~~ then it is nlw id of full nlw or nlw id for first subnet.

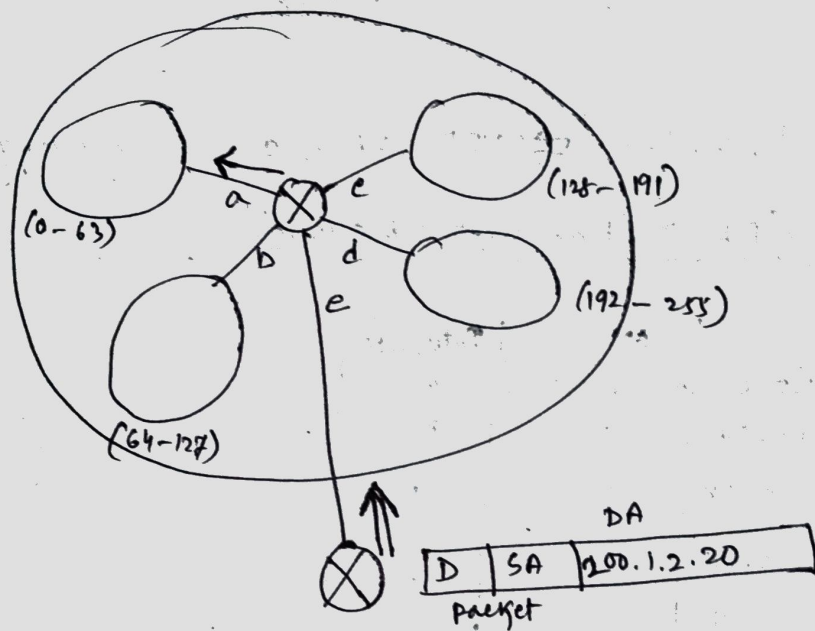
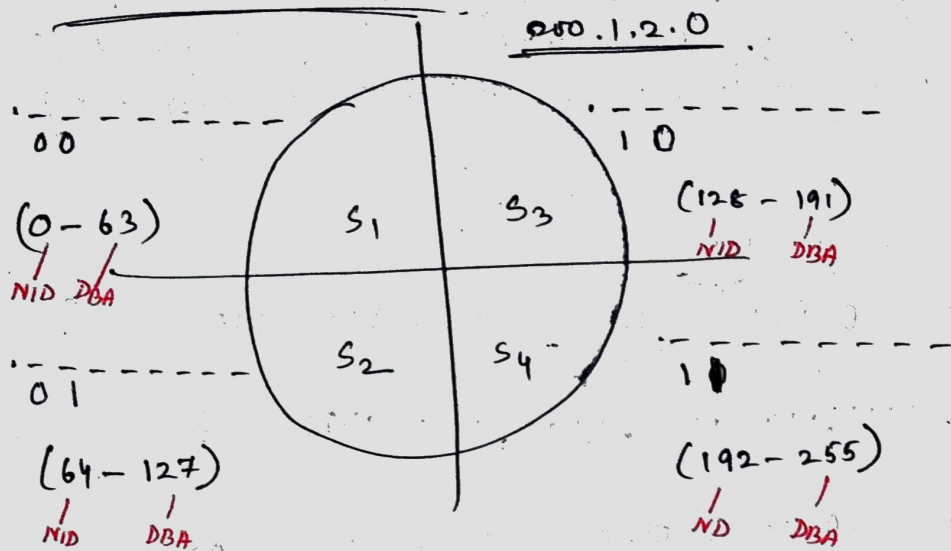
$\Rightarrow$ Same as when we use $200.1.2.$ ~~127~~ ~~0~~ 255 then it is DBA of full nlw or DBA for second subnet.

$\Rightarrow$ We solve this problem according to perspective, means if we are present inside the nlw then we know about the subnet then it is subnet id or DBA of second subnet & if we are outside the nlw then we don't know about the subnet then it is nlw id or DBA of full nlw.

$\Rightarrow$ Like



Divide entire n/w in four part



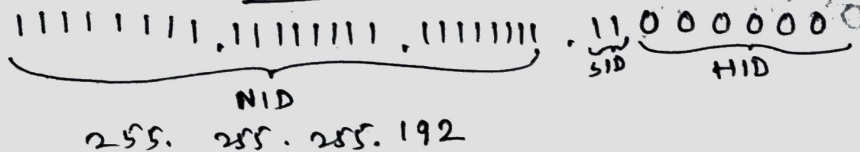
⇒ Now here how the internal router identifies the n/w where this ip address is belong?

⇒ By using subnet mask (32bit)

All 1's ⇒ NID & SID

All 0's ⇒ HSD

⇒ Subnet mask for 200.1.2.0



⇒ Advantage of subnet mask is: given any ip address if you take the subnet mask & you bitwise AND it then you are going to get a output the n/w id of the n/w to which this ip address belong.

Bx

IP add. 200.1.2.130

11001000.00000001.00000010.10000010

Subnet mask 11111111.11111111.11111111.11000000

Bitwise AND 11001000.00000001.00000010.10000000
 (AND 200 only
 any no. with 255 we get that no.) 1 . 2 . 128

200.1.2.128 ⇒ NID where 200.1.2.130 belong.

Routing Table (FLSM (Fixed Length Subnet Mask))

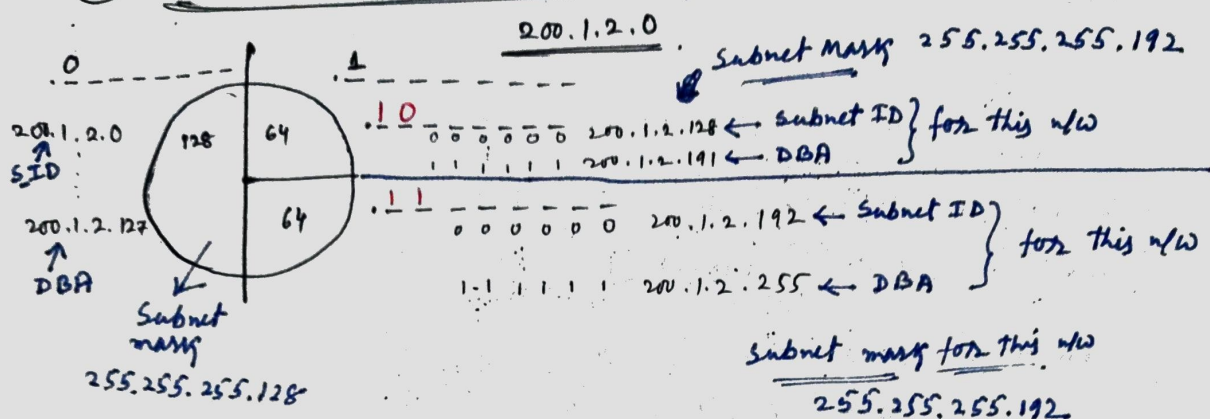
NID	SM	Interface
200.1.2.0	255.255.255.192	a
200.1.2.64	"	b
200.1.2.128	"	c
200.1.2.192	"	d
0.0.0.0	0.0.0.0	e

default entry

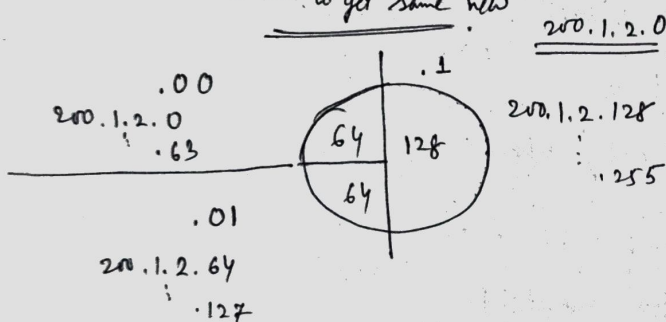
⇒ When we find exact match we send that packet to its corresponding interface.

⇒ Some time we find more than one match, in that case choose the interface which have longest subnet mask (longest means no. of 1's is ~~larger~~ more than other).

L4 Variable length subnet masking (VLSM)



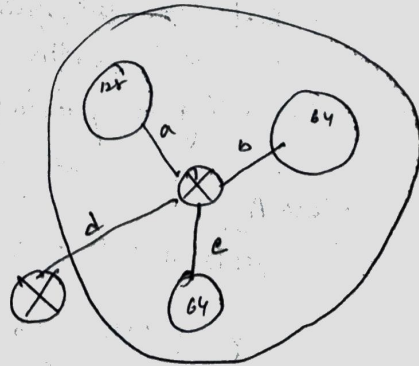
Another Ex. to get same n/w



Subnet mask are same as previous one.

For first ex.

NID	SM	Interface
200.1.2.0	255.255.255.128	a
200.1.2.128	255.255.255.192	b
200.1.2.192	255.255.255.192	c
0.0.0.0	0.0.0.0	d



For this subnet mask what we can understand?

255.255.255.192
11111111.11111111.11111111.11000000

So in 8M no. of 1's = N/W ID + Subnet ID

and no. of 0's = Host ID

So here, NID + SID = 26 1's

HID = 6 0's

⇒ So from subnet mask we can directly find out the number of IP address are possible in each subnet.

⇒ In this example 6 0's that is

$$2^6 = 64 \text{ IP add. / subnet}$$

↓

$$2^6 - 2 = 64 - 2 = 62 \text{ host}$$

→ One ip add. for subnet ID

→ one ip add. for DBA

⇒ But we can^{not} find individually the NID & SID. For that purpose we want another information, i.e. which class it is belong?

According to previous example:

class A.

$$\text{NID} + \text{SID} = 26$$

$$8 + \text{SID} = 26$$

$$\text{SID} = 18$$

So 2^{18} subnets are possible

class B.

$$\text{NID} + \text{SID} = 26$$

$$16 + \text{SID} = 26$$

$$\text{SID} = 10$$

So 2^{10} subnets are possible

class C

$$\text{NID} + \text{SID} = 26$$

$$24 + \text{SID} = 26$$

$$\text{SID} = 2$$

So 2^2 subnets are possible

Note:

* $\rightarrow$ If $SID = 3$ then how many subnet are possible?

$2^3 = 8$ but it have some confusion.

$\left(\begin{matrix} 3 \\ 2-2 \end{matrix} \right) \rightarrow \text{subnet}$
 $\quad \quad \quad \text{or}$
 $\left(\begin{matrix} 3 \\ 2 \end{matrix} \right) \rightarrow \text{subnet}$

Basically both are possible

$\rightarrow$ Previously they did not allow to use all 0's & all 1's in the subnet ID part, which means they use only $(2^3 - 2)$ subnet but currently they are using all the subnet i.e. (2^3) .

$\rightarrow$ In CISCO Router, have a command called "subnet 0", using that command we can use all the subnet.

* $\rightarrow$ Any practical use subnet mask will contain a sequence of 1's followed by a sequence of zeros.

$\rightarrow$ But theoretically it is possible that instead of choosing the SID part from the beginning you can choose from anywhere. like

... .. 11000000

Question like:

255.255.255.15 it is SM then how many subnet are possible for class C?

count the no. of 1's

255 . 255 . 255 . 15

no. of 1's = $8 \cdot 8 \cdot 8 \cdot 4 = 28$ 1's

so $NID + SID = 28$

$24 + SID = 28$

$SID = 4$

so no. of subnet possible is $= 2^4 = 16$

but when we check the subnet mask

11111111.11111111.11111111.00001111

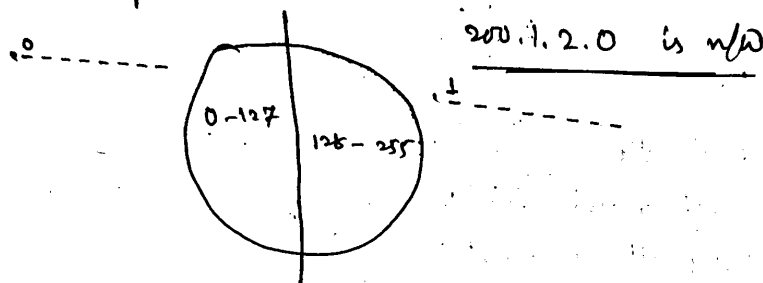
255 . 255 . 255 . 15

that is not a practically use subnet mask but theoretically we can use it.

⇒ The Reason is that, means why we not taken 8th significant bit for subnet ID part?

⇒ Because we not be able to divide the subnet into ranges.

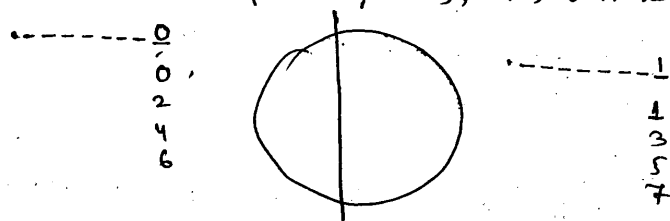
⇒ For example: If i want to divide a subnet into two part,



then how many bit we need from host part?

⇒ One bit

⇒ do from 8 bit if we choose one bit from 1st significant site, then it will be like this, means 0 IP add. ^{fall} in one subnet & 1 IP add.



fall in another subnet so all even no. IP add. in one subnet & all odd no. IP add in another subnet it is not practically possible so always we use most significant bit for choosing subnet id part.

25

Subnet Masking Question

	Default SM	Subnet Mask	No. of Hosts	Subnets in class A	Subnets in class B	Subnets in class C	2 ¹⁰ / 10
00000000 - 0	→	255.0.0.0	2 ²⁴ - 2	2 ⁸ = 1 (no bit to create subnets in subnet)	—	—	—
10000000 - 128		255.128.0.0	2 ²³ - 2	2 ⁷	—	—	—
11000000 - 192		255.192.0.0	2 ²² - 2	2 ⁶	—	—	1
11100000 - 224		255.240.0.0	2 ²⁰ - 2	2 ⁴	—	—	2 ²
11110000 - 240		255.255.0.0	2 ¹⁶ - 2	2 ⁸	2 ⁸ = 1	—	2 ⁶
11111000 - 248		255.255.254.0	2 ⁹ - 2	2 ¹⁵	2 ⁷	—	2 ³
11111100 - 252		255.255.255.0	2 ⁸ - 2	2 ¹⁶	2 ⁸	1	2 ¹⁴
11111110 - 254		255.255.255.224	2 ⁵ - 2	2 ¹⁹	2 ¹¹	3 2	2 ¹⁷
11111111 - 255		255.255.255.240	2 ⁴ - 2	2 ²⁰	2 ¹²	4 2	2 ¹⁸

⇒ In exam they can give subnet mask & ip address from that you could find out the class.

like:

255.255.255.192 ← SM

200.1.2.3 ← IP add.

So from IP add we understand that it is class C.
 & from SM we ~~also~~ find that no. of host is

$$(2^6 - 2) \text{ host / subnet}$$

and no. of subnet is

$$2^2 \text{ [because it is class C]}$$

$$\text{NID} + \text{SID} = 26$$

$$24 + \text{SID} = 26$$

$$\text{SID} = 2$$

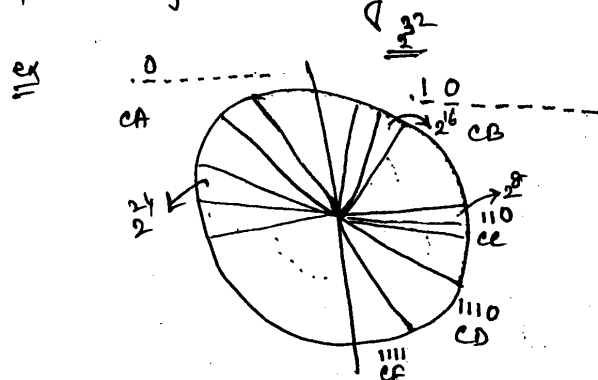
$$\text{so no. of subnet is } 2^2 = 4$$

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L6

classless Inter Domain Routing (CIDR)

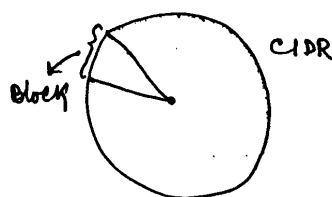
⇒ In classfull addressing have some disadvantages.



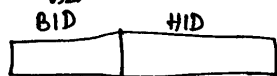
IANA (Internet assigned number authority)
Having the control to manage all the IP addresses.

⇒ Problem is flexibility is very poor, because suppose we want 100 IP add. for that we buy class C IP add, that provide (2^8) 256 host, but we want only 100 IP add or if we want 300 IP add. then we buy class B IP add. that provide (2^{16}) nearly 64 thousand host, so many of the IP add. is wasted.

⇒ Still 1990's people used this classfull classification. But it not flexible then people come up with classless IP addressing.



⇒ The main disadvantage of CIDR rule is in classfull add. if we know the class then easily identify the NID & HID part, but in CIDR it is difficult to find out the NID or BID (Block ID) & HID part.



a.b.c.d/n
↓
It is dotted decimal notation
↓
It is no. of bit in NID or BID part

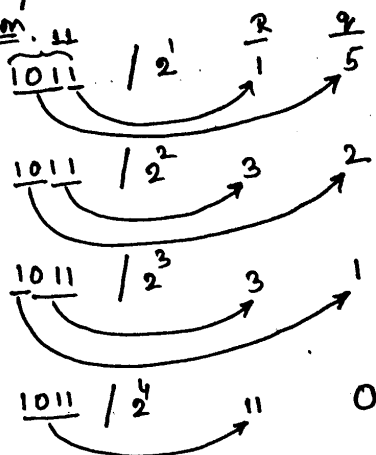
⇒ ex 20.18.50.100/20 ⇒ Its mean, In the block in which this IP add. is present, the number of bit is used for NID or BID is 20.

⇒ So if NID = 20
HID = 12 (32-20)
no. of IP add. is 2^{12}

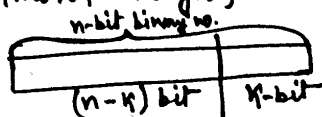
Rules for CIDR blocks

- ① All IP add. should be contiguous
- ② Block size always be a power of 2 (2^n). [Then easily divide BID & HID part]

Reason



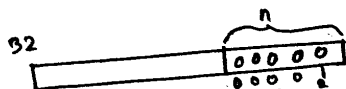
So if I have a n -bit binary number (it should be binary) & if I divide it by 2^i then remainder (R) is nothing but least significant i -bit, & most significant $(n-i)$ bit is quotient (Q).



And if it is decimal number then dividing by 10 is easy

1234	$/ 10^1$	$\Rightarrow$	R	Q
			4	123
	10^2	$\Rightarrow$	34	12
	10^3	$\Rightarrow$	234	1
	10^4	$\Rightarrow$	1234	0

- ③ First IP address in the block should be evenly divisible by size of the block.



If first IP add. divisible by size of the block (i.e. 2^n) then least significant bit is n bit, that all are 0's (because it is evenly divisible), so starting IP add having all 0's in least significant bit & we can use this add. as BID & also we get all (2^n) IP add. if we start from all 0's.

⇒ ^{CIDR} Using this rule now we can check the given IP address is valid or not.
CIDR block or not.

BID ← 100.1.2.32
100.1.2.33
⋮
DBA ← 100.1.2.47

$$\text{Block Size} = 2^4 = 16$$

$$HID = 4 \text{ bit}$$

$$\text{So, } NID = 32 - 4 = 28 \text{ bit}$$

So, we can represent this block by using /28 with any of the IP address like $100.1.2.32/28$ ← Or define whole block also

- ① Rule 1 is satisfied because all IP address are contiguous (32-47).
- ② Rule 2 is satisfied because size of the block (32-47 is 16) that is power of 2 = 2^4 (16).
- ③ Now check Rule 3, if we divide first IP address by 2^4 then least significant bit is 4 & if least significant bit is all 0's then rule 3 is satisfied.

100.1.2.00100000

least significant bit is all 0's

So Rule 3 is also satisfied, so this given block is also valid CIDR block.

⇒ ^{ex} BID ← 20.10.30.32
20.10.30.33
⋮
DBA ← 20.10.30.63

① Rule 1 ✓

② Rule 2 ✓ ($32 = 2^5$)

③ Rule 3 ✓ (all 0's 00100000)

$$NID = 32 - 5 = 27 \text{ bit}$$

20.10.30.32/27 or 20.10.30.33/27

⇒ ^{ex} BID ← 150.10.20.64
150.10.20.65
⋮
DBA ← 150.10.20.127

① Rule 1 ✓

② Rule 2 ✓ ($64 = 2^6$)

③ Rule 3 ✓ (all 0's 01000000)

$$NID = 32 - 6 = 26 \text{ bit}$$

150.10.20.64/26 or
150.10.20.66/26 etc.

⇒ Given CIDR representation how we get the entire block?

ex
20.10.30.35/27

Ans

20.10.30.00100011

BID (27 bit)
HID (5 bit)

→ put all 0's in HID part to get first IP add. of the block.
 → put all 1's in HID part to get last IP add. of the block.

BID →	20.10.30.00100000	← 32
	20.10.30.00100001	← 33
	⋮	
DRA →	20.10.30.00111111	← 63

ex
100.1.2.35/28

Ans

100.1.2.00100011

BID (28 bit)
HID (4 bit)

BID →	100.1.2.00100000	← 32
	100.1.2.00100001	← 33
	⋮	
DRA →	100.1.2.00101111	← 47

ex
100.1.2.35/20

Ans

100.1.00000100	00100011
----------------	----------

BID (20 bit)
HID 12 bit

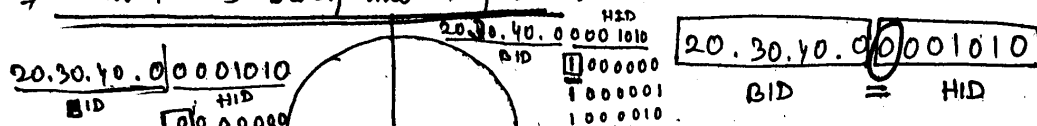
100.1.00000000.00000000	→ 100.1.0.0	← BID
100.1.00000000.00000001	→ 100.1.0.1	
⋮	⋮	
100.1.00001111.11111111	→ 100.1.15.255	← DRA

L7 Subnetting in CIDR, VLSM in CIDR

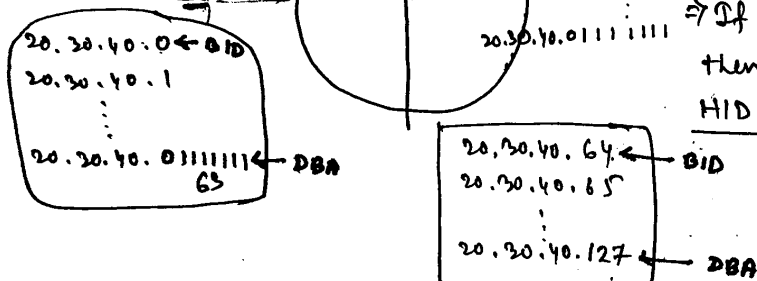
Subnetting in CIDR

20.30.40.10/25

⇒ I want this block into 2 part :



⇒ If I divide this block into 2 part then we can choose one bit from HID part.



Now bit of BID = BID + subnet ID
= 25 + 1 = 26

same as

20.30.40.64/26

Now represent IP add. like

20.30.40.0/26

It also represent subnet mask.

subnet mask represent 26 1's followed by 6 0's

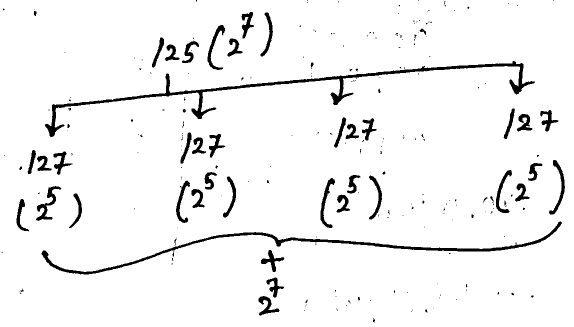
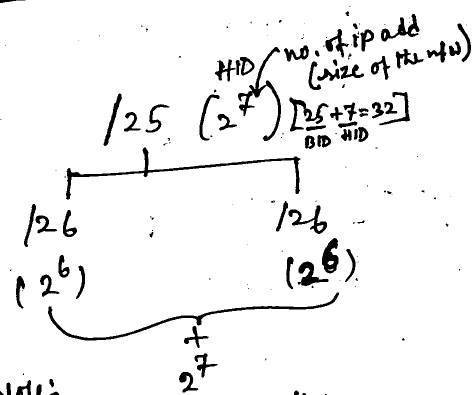
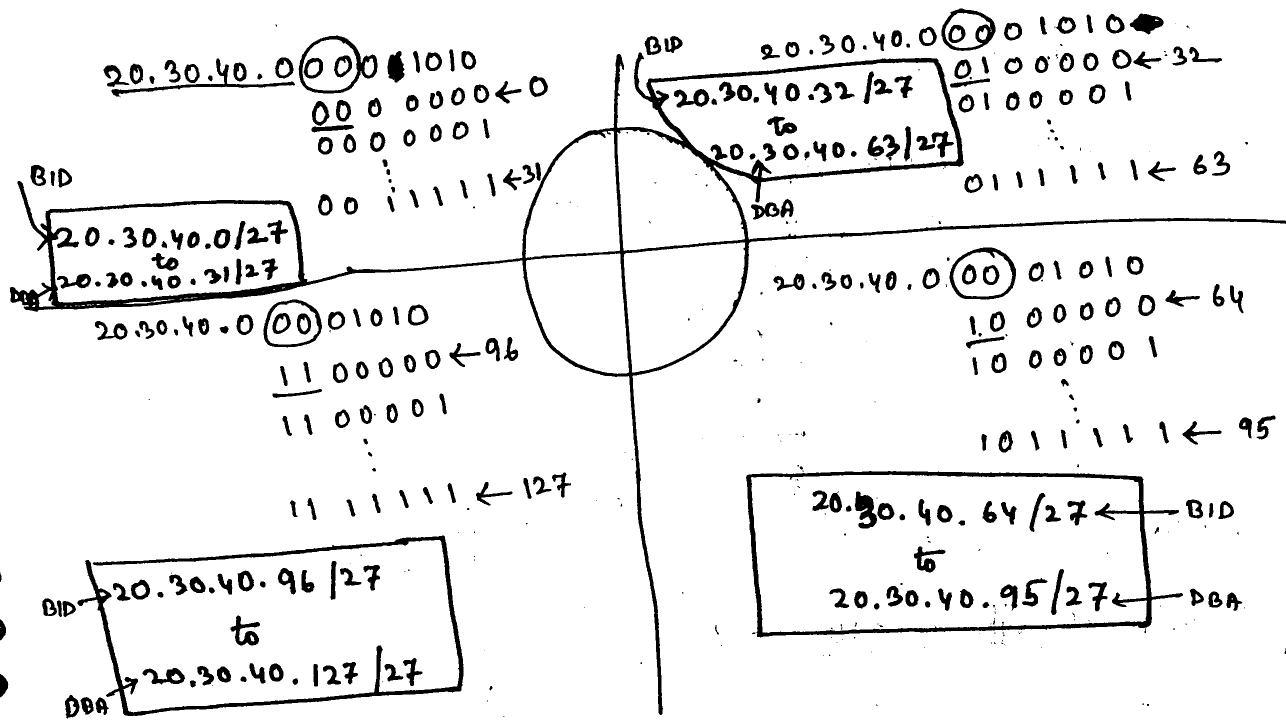
20.30.40.10/25

⇒ I want this block into 4 part

20.30.40.0 | 0001010
BID HID

□ □ - - - -

↓
This two bit are chosen for divide this block into 4 part



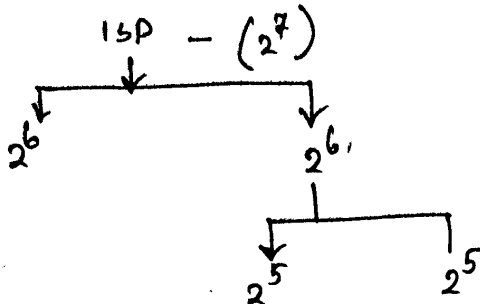
Note: do after dividing the n/w into two part the size of the entire n/w remains same & it also happen ~~when~~ when we divide the n/w into 4 part.

NLSM in CIDR blocks

ex

20.30.40.10/25

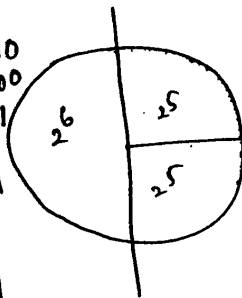
⇒ We divide this block into 3 part like this:



one solution

20.30.40.0/0001010

20.30.40.00001010
0000000
0000001
...
0111111



20.30.40.00001010
1000000
1000001
1011111

20.30.40.64/27
to
20.30.40.95/27

20.30.40.0/26
to
20.30.40.63/26

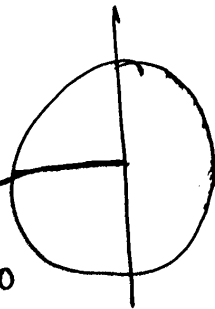
20.30.40.00001010
1000000
1100001
...
1111111

20.30.40.96/27
to
20.30.40.127/27

Another solution

20.30.40.0/27
to
20.30.40.31/27

20.30.40.32/27
to
20.30.40.63/27

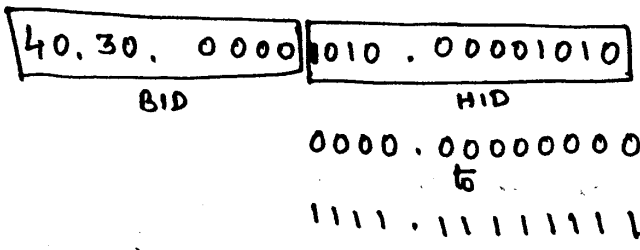


01
01000000
...
01111111

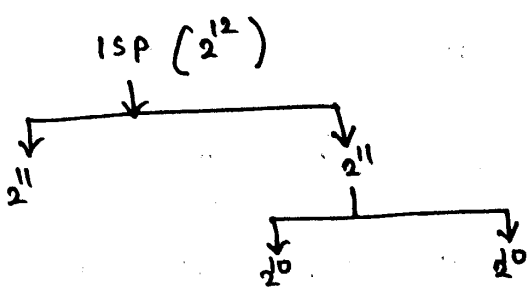
20.30.40.64/26
to
20.30.40.127/26

Q.

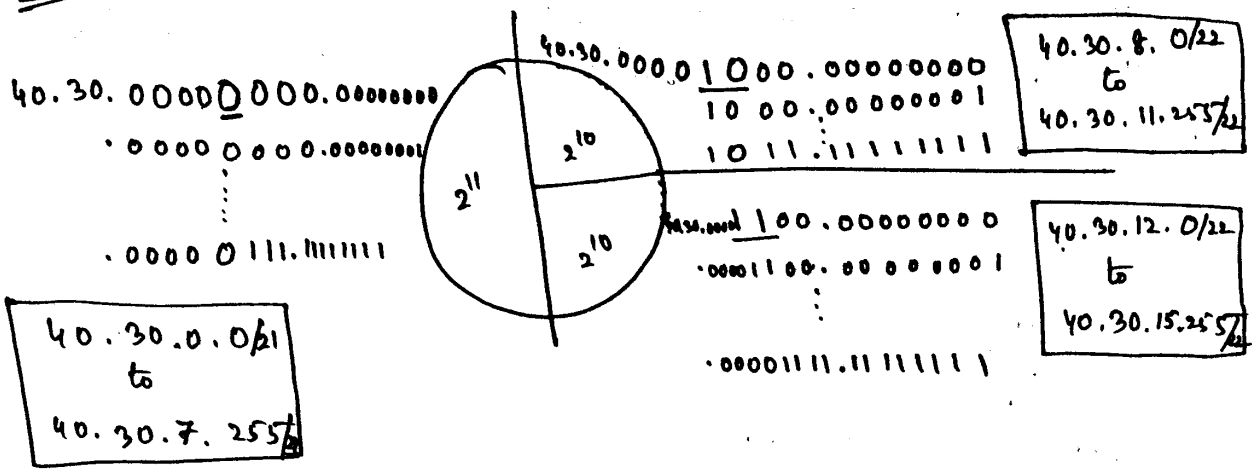
40.30.10.10/20



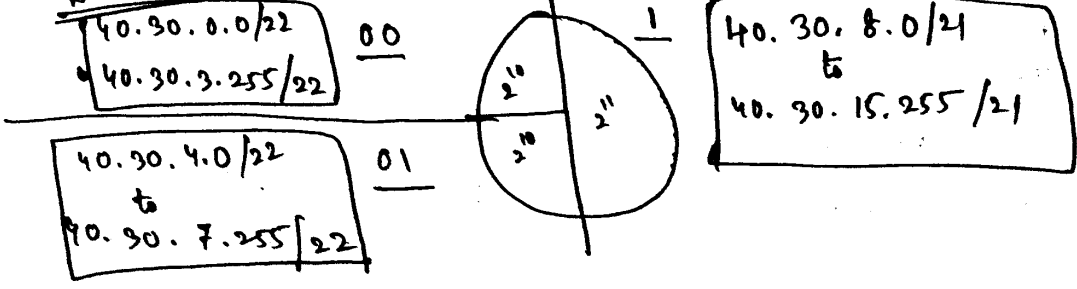
So First IP add = 40.30.0.0
Last IP add = 40.30.15.255 } Range of the block



one sub

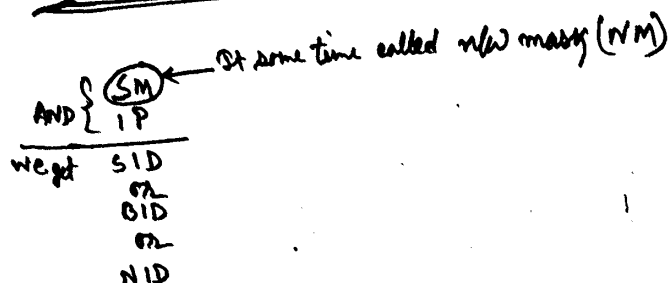


Another sub



(L8)

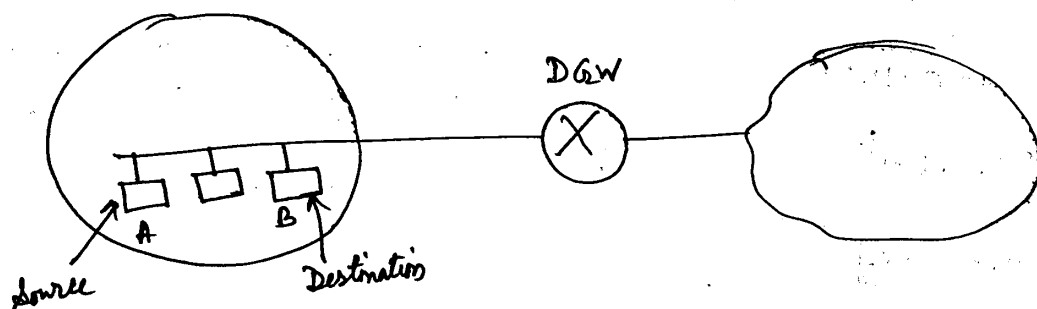
Some interesting problems on subnet mask



When you connect your computer to the internet then ISP (Internet Service Provider) provides something to you:

- IPv4 (IP add.) : Own IP add.
- DGW (Default Gateway) : It is nothing but Default Router
- SM (Subnet mask) : SM you should use
- DNS (Domain name system) : which is used to convert the Domain Name into IP address.

→ We know the use of IPv4, DGW, DNS but why we use SM, because SM is used by ^{the} router to find the new of destination IP add, if we are not a Router?



Host A: I_A
: SM_A

Host B: I_B

Suppose A knows IP add of B. Now if B is in same new where A present then A directly send the msg to B otherwise it send the msg to DGW then from DGW to send to destination.

So how to find the • NID of B .

First
AND { I_A
 SMA

Weight NID of A
 respect to A

AND { I_B
 SMA

Weight NID of B
 respect to A

Then if both are
equal then A & B are
in same n/w otherwise
not.

That is the reason to use SM & ISP provides
SM to you.

Ex

A : I_A : 200.1.2.10

B : I_B : 200.1.2.130

SMA : 255.255.255.128

AND { 200.1.2.00001010
 255.255.255.10000000

200.1.2.00000000

200.1.2.10000010
255.255.255.10000000

200.1.2.10000000

So NID of A is 200.1.2.0

So NID of B is 200.1.2.128

So NID of A & B are not same

Some time SM create elusion that even tho A & B is in same n/w
but they can provide info to show that they are in different n/w.

Ex

A : I_A : 200.1.2.10

B : I_B : 200.1.2.126

SMA : 255.255.255.128

AND
NID = 200.1.2.0

01111110
10000000

11111110

NID of B = 200.1.2.0

Pr A

$I_A: 200.1.2.10$

$SM_A: 255.255.255.128$

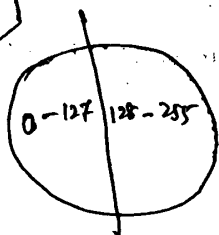
$I_B: 200.1.2.69$

$SM_B: 255.255.255.192$

$\frac{I_A}{SM_A}$
 $200.1.2.0$
 MID of A according to A

$\frac{I_B}{SM_A}$
 $200.1.2.0$
 MID of B according to A

So A thinks that
 A & B are in same
 n/w

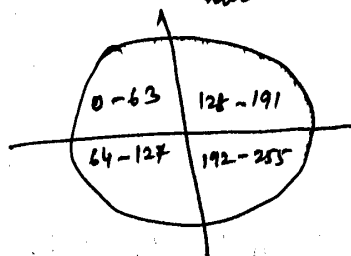


So here (in A) 10 & 69
 is in same n/w

$\frac{I_B}{SM_B}$
 $200.1.2.64$
 MID of B according to B

$\frac{I_A}{SM_B}$
 $200.1.2.0$
 MID of A according to B

So B thinks that
 A & B are in different
 n/w



here (in B) 10 & 69 is in
 different n/w

⇒ subnet mask is just a perception, it is not reality

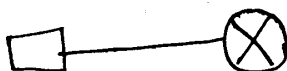
⇒ It provides security

⇒ In your computer if you type IP config then you see that

$SM: 255.255.255.255$

why they give it?

If you take a connection in your modem (Airtel or reliance)
 then you are having a Point to Point connection with your router



Even if any host of your n/w, then also you send the packet
 through router, you can't send packet directly, because of this SM.

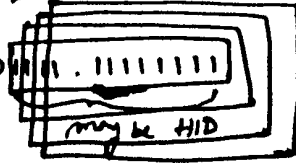
if any no. AND with 255, will produce that no. only.
 if do if my IP add AND with 255.255.255.255 then we
 get a NID that is nothing but my own IP add. (IA = NID)

(*) Given a DBA, possible to find NID or SM?

DBA: 200.1.15.255

[Don't think that it is class C.
 It is class-less classification]

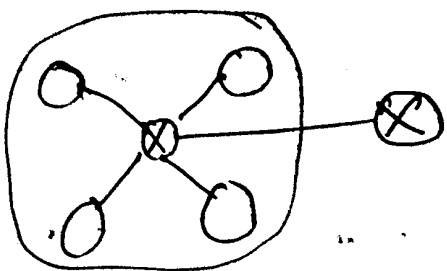
11001000.00000001.0000



/20
 /21
 /22
 ... } There could be many answer

29 Supernetting or aggregation

(one entry every n/w)
 if more n/w then routing table entry is more. so if n/w is less
 then routing table entry is less, so execution process more faster.



Instead of 4 we can give only one entry.

Supernetting is not combining the n/w, it just perception.

US → India
 ↓
 US need not having
 the database of
 cities or states etc.

⇒ like CIDR rule there ~~are~~ some rule's using ~~as~~ which we have to apply Aggregation.

⇒ That means all the n/w are not suitable for Aggregation.

Rule (Same as CIDR rule)

(row 2)

⇒ All the n/w should be contiguous.

⇒ Size of all the n/w should be same & moreover they should be of 2^n .

⇒ First n/w ID divisible by size of the block (2^n)
(means total size of the ~~block~~ supernet)

① 200.1.0.0/24

② 200.1.1.0/24

③ 200.1.2.0/24

④ 200.1.3.0/24

① This are contiguous

② (2^8) that are same size

③ 1st IP add are divisible by total supernet.

$$\text{Total supernet} = 4 \times 2^8 = 2^{10}$$

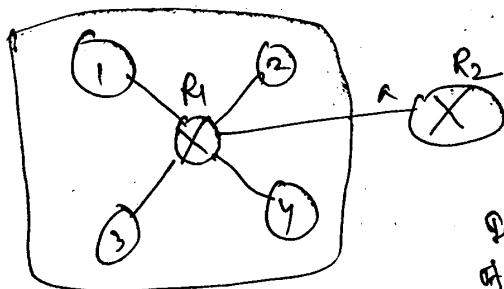
200.1.0.0 $8/2^{10}$

200.1.00000000.00000000

↓
Remainder (least significant bit)
that is '0'

so first IP add. divisible by 2^{10}

so it is valid n/w's that are possible for aggregation.



Instead of this four entry we ~~can~~ give one entry that is supernet ID.

NID	SM	Outgoing
200.1.0.0	255.255.255.0	a
200.1.1.0	"	"
200.1.2.0	"	"
200.1.3.0	"	"

Supernet Mask (32 bit)

All 1's = Fixed Part
All 0's = Variable part

Fixed part	variable part
200.1.00000000	00.00000000
200.1.00000000	01.00000000
200.1.00000000	10.00000000
200.1.00000000	11.00000000
255.255.11111111	00.00000000

255.255.252.0 $\leftarrow$ Supernet mask
(22 1's)

IP AND Supernet mask = Supernet ID

Ex

$$\begin{array}{r} 200.1.0.0 \\ 255.255.252.0 \\ \hline 200.1.0.0 \text{ Supernet ID} \end{array}$$

Short Cut

$\Rightarrow$ If all new host ID is 2^8
then $4 \times 2^8 = 2^{10}$

so supernet mask = $\left[\frac{32-10}{2} = 22 \right]$

$\therefore$ first IP add of the n/w is supernet ID . i.e.

200.1.0.0

Routing Table

ND	SM	Interface
200.1.0.0	255.255.252.0	a

Ex

supernet ID $\rightarrow$ 20.1.32.0/24
 20.1.33.0/24
 :
 20.1.47.0/24

① ✓

② ✓ (2^8)

③ $16 \times 2^8 = 2^{12}$ (size) ✓

20.1.32.0

20.1.00100000.00000000
 Remainder 0

/20 Supernet mask
 255.255.240.0 Supernet mask

Ex

100.1.2.0/25

supernet ID $\rightarrow$ 100.1.2.128/26 } size (2^6)
 100.1.2.192/26 } so $2 \times 2^6 = 2^7$

100.1.2.10000000
 Remainder 0

Supernet mask /25

supernet ID $\rightarrow$ 100.1.2.0/25 } size (2^7)
 100.1.2.128/25 } so $2 \times 2^7 = 2^8$

100.1.2.00000000
 Remainder 0

supernet mask /24

so
 100.1.2.0/24